

OPEN-SET METRIC LEARNING FOR PERSON RE-IDENTIFICATION IN THE WILD

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ABSTRACT

Person re-identification in the wild needs to simultaneously (frame-wise) detect and re-identify persons and has wide utility in practical scenarios. However, such tasks come with an additional open-set re-ID challenge as all probe persons may not necessarily be present in the (frame-wise) dynamic gallery. Traditional or close-set re-ID systems are not equipped to handle such cases and raise several false alarms as a result. To cope with such challenges open-set metric learning (OSML), based on the concept of Large margin nearest neighbor (LMNN) approach, is proposed. We term our method Open-Set LMNN (OS-LMNN). The goal of separating impostor samples from the genuine samples is achieved through a joint optimization of the Weibull distribution and the Mahalanobis metric learned through this OS-LMNN approach. The rejection is performed based on low probability over distance of imposter pairs. Exhaustive experiments with other metric learning techniques over the publicly available PRW dataset clearly demonstrate the robustness of our approach.

Index Terms— Open-set metric learning, Person re-identification in Wild, LMNN, Weibull distribution.

1. INTRODUCTION

Identifying target persons across non-overlapping views is an important retrieval task for potential surveillance applications. Such a task is known as person re-identification (re-ID). The problem is inherently challenging [1] as interest person can undergo drastic variations in illumination, viewpoint, occlusion and background clutter. Several existing methods [2–4] have addressed this re-ID problem assuming that query/probe person is always present in the search list (gallery set). This close-set assumption is not realistic while performing re-ID over long duration videos. Instead, re-ID over open set (i.e., absence of probe in gallery set) becomes important in many practical scenarios. Furthermore, current re-ID methods are likely to perform below par in such open settings. Thus, a dedicated re-ID system [5] capable of person detection as well as re-identification is surely more desirable.

Person re-identification in the wild is another desirable feature of any efficient re-ID system. Zheng *et al.* [6] first

performed re-identification in the wild by detecting an re-identifying persons in a single framework. Unlike Zheng, earlier works [1, 4] usually performed re-ID over pre-extracted bounding boxes which are either human annotated [7] or pre-detected [8]. Re-identification in the wild points towards a generalized re-ID task [9, 10] and is easily deployable in any practical scenarios with minimal adjustments.

Re-identification in the wild is also allied to open set person re-identification [5, 11] when performed frame-wise. Clearly, all the probe persons are not present in a particular frame. This situation arises especially when a probe set is large in comparison to dynamically (frame-wise) generated gallery. A rejection mechanism is necessary in such cases for reduced false alarm. Existing metric learning algorithms [12–15] are unequipped to deal with such situations. To deal with this limitation, we have developed open-set metric learning (OSML), a new metric learning approach, based on the concept of open set recognition [16] and Large margin nearest neighbor (LMNN) approach [12]. We term our method open-set LMNN (OS-LMNN).

The contributions of our work are the following: (a) Theoretically speaking, we have introduced a new concept of open set metric learning (OSML) for a more realistic open set re-ID setting. In this approach, we jointly learn the Mahalanobis metric using the proposed OS-LMNN along with a Weibull distribution. The Weibull distribution enforces better separation of imposter samples from the distribution of genuine samples during optimization. Furthermore, the learned Weibull distribution rejects obvious imposter samples before ranking thereby leading to a better re-ID. (b) Dynamically (frame-wise) generated gallery is an open set relative to the probe set. So, from the application perspective, we have essentially extended re-ID in the wild task to open set re-id settings by detecting and performing re-ID frame-wise.

2. PROPOSED METHOD

In this section, we formally present the problem definition, theoretical foundations and the proposed framework.

2.1. Problem Definition

Given a person from probe set P , the objective is to perform re-ID over gallery set G of pedestrians detected in Wild [6]. The set G is generated frame-wise (dynamic). Re-identifying person in a frame is an open set problem [5] as his/her presence

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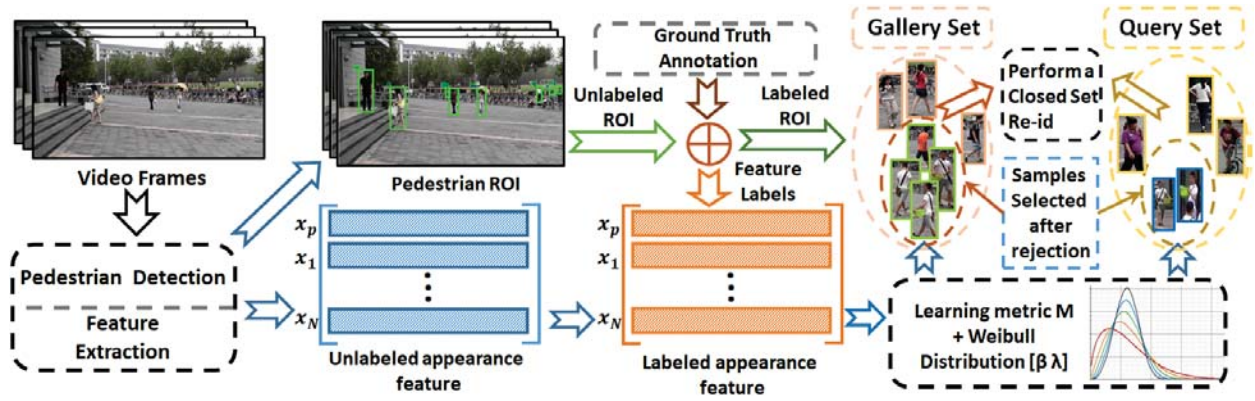


Fig. 1. An illustrative overview of the proposed solution

is not always ensured. Our re-ID problem can be divided into two sub-tasks: (a) whether a given probe has a true match in dynamic gallery set, and (b) if yes, which one is the correct match.

2.2. OS-LMNN Approach

The trivial way of addressing the above problem is by rejecting samples (close set conversion) using a heuristic threshold [5] over similarity measure. While based on the concept of Extreme Value Theorem (EVT), some recent works [16, 17] have shown that a learned Weibull distribution can better represent unlikely samples in the tail of the distribution. Thus such samples can be easily separated using a smaller threshold over probability. In our proposed approach, the Weibull distribution is learned along with Mahalanobis metric by alternatively optimizing a regularized error function.

Let x_i and x_j be the feature representation of probe sample $i \in P$ and gallery sample $j \in G$. The distance $D_{ij}^{\mathbf{M}}$ between samples i and j using Mahalanobis metric $\mathbf{M}$ can be represented as:

$$D_{ij}^{\mathbf{M}} = (x_i - x_j)^T \mathbf{M} (x_i - x_j) \quad (1)$$

The metric $\mathbf{M}$ in above equation can be learned via LMNN [12] approach by optimizing the following error function:

$$\epsilon(\mathbf{M}) = (1 - \mu) \sum_{i \rightsquigarrow j} D_{ij}^{\mathbf{M}} + \mu \sum_{i, j \rightsquigarrow i} \sum_k [\alpha + D_{ij}^{\mathbf{M}} - D_{ik}^{\mathbf{M}}]_+ \quad (2)$$

where, the first (pull) error term corresponds to distance between similar pairs (i, j) . The second (push) error term corresponds to distance margin (α) that an imposter (k) sample can intrude a true (j) sample with respect to an anchor (i). All such combinations of anchor (i), true (j) and imposter (k) set is a valid triplet and $[\cdot]_+ = \max(\cdot, 0)$ represents standard hinge loss function. Here, μ is a weighting parameter that balances the push and pull factors. Usually, μ is fixed at a particular value. In sharp contrast, we dynamically adjust this weight to better separate imposters from the distribution of genuine samples. The dynamic weight is computed based on the learned Weibull distribution [16]. The standard probability density function (pdf) of Weibull distribution can be

represented as:

$$\rho(x; \beta, \lambda) = \begin{cases} \frac{\beta}{\lambda} \left(\frac{x}{\lambda}\right)^{\beta-1} e^{-(x/\lambda)^\beta} & x \geq 0 \\ 0 & x < 0 \end{cases} \quad (3)$$

where, $\beta > 0$ and $\lambda > 0$ are respectively the shape and the scale parameter of Weibull distribution [16]. The cumulative distribution function (CDF) of Weibull distribution can be represented as:

$$F(x; \beta, \lambda) = [1 - e^{-(x/\lambda)^\beta}] \quad (4)$$

where, $F \in [0, 1]$ is a bounded monotonically increasing function. Utilizing these characteristics of Weibull CDF we jointly learn Metric along with Weibull distribution based on our proposed error function:

$$\epsilon(\mathbf{M}, \beta, \lambda) = \sum_{i, j \rightsquigarrow i} \sum_k \left\{ \left(\frac{\omega_{ki}}{1 + \omega_{ki}} \right) \cdot D_{ij}^{\mathbf{M}} + \left(\frac{1}{1 + \omega_{ki}} \right) \cdot [\alpha + D_{ij}^{\mathbf{M}} - D_{ik}^{\mathbf{M}}]_+ \right\} \quad (5)$$

where, $\omega_{ki} = F(D_{ki}^{\mathbf{M}}; \beta, \lambda)$ (see equation 4) and μ_j is mean of samples of person j belonging to same ID. Based on Weibull CDF, smaller distance between dissimilar pairs decreases ω_{ki} thereby increasing the push factor weight $1/(1 + \omega_{ki})$ with respect to the pull one and vice-versa. Weibull parameters $\mathbf{w} = [\beta, \lambda]$ learned based on equation 5 will saturate to an arbitrary high value. In order to restrict such convergence, we regularize the above error function as:

$$\mathbf{M}^*, \beta^*, \lambda^* = \underset{\mathbf{M}, \beta, \lambda}{\operatorname{argmin}} [\epsilon(\mathbf{M}, \beta, \lambda) + \gamma \cdot \mathcal{R}(\beta, \lambda)] \quad (6)$$

Here, $\mathcal{R}(\beta, \lambda) = \frac{1}{2} \mathcal{N} \cdot (\beta + \lambda)$ is a regularization term and $\mathcal{N}$ is the total number of valid triplets. We have adopted L-BFGS-B [18] optimizer to solve the objective $J = \epsilon(\mathbf{M}, \beta, \lambda) + \gamma \cdot \mathcal{R}(\beta, \lambda)$ (equation 6) via alternatively fixing $\mathbf{M}$ and $\mathbf{w}$. The gradient of J with respect to $\mathbf{M}$ can be computed as:

$$\frac{\partial J}{\partial \mathbf{M}} = \sum_{i, j \rightsquigarrow i} \sum_k \left\{ \left(\frac{\omega_{ki}}{1 + \omega_{ki}} \right) \cdot \mathbf{C}_{ij} + \left(\frac{1}{1 + \omega_{ki}} \right) \cdot (\mathbf{C}_{ij} - \mathbf{C}_{ik}) + \left(\frac{1}{1 + \omega_{ki}} \right)^2 \cdot \rho(D_{ki}^{\mathbf{M}}; \mu_j) \cdot (D_{ij}^{\mathbf{M}} - [\alpha + D_{ij}^{\mathbf{M}} - D_{ik}^{\mathbf{M}}]_+) \cdot \mathbf{C}_{\mu_i, k} \right\} \quad (7)$$

where, $\mathbf{C}_{ij} = (x_i - x_j)(x_i - x_j)^T$ is the outer product of sample x_i and x_j and $\mathbf{C}_{\mu_i, k} = (\mu_i - x_k)(\mu_i - x_k)^T$ is the outer product of μ_i and x_k . The gradient of J with respect to $\mathbf{w}$ is:

$$\frac{\partial J}{\partial \mathbf{w}} = \begin{cases} \frac{1}{(1+\omega_{ki})^2} \cdot \frac{\partial \omega_{ji}}{\partial \mathbf{w}} \sum_{i,j \rightsquigarrow i,k} D_{ij}^{\mathbf{M}^*} + \frac{1}{2} \gamma \cdot \mathcal{N} & \varepsilon_{push} \leq 0 \\ \frac{1}{(1+\omega_{ki})^2} \cdot \frac{\partial \omega_{ji}}{\partial \mathbf{w}} \sum_{i,j \rightsquigarrow i,k} [D_{ik}^{\mathbf{M}^*} - \alpha]_+ + \frac{1}{2} \gamma \cdot \mathcal{N} & \varepsilon_{push} \geq 0 \end{cases} \quad (8)$$

where, $\varepsilon_{push} = [\alpha + D_{ij}^{\mathbf{M}} - D_{ik}^{\mathbf{M}}]_+$ and

$$\frac{\partial \omega_{ji}}{\partial \mathbf{w}} = \begin{bmatrix} \frac{\partial \omega_{ji}}{\partial \beta} \\ \frac{\partial \omega_{ji}}{\partial \lambda} \end{bmatrix} = \begin{bmatrix} -\frac{\beta}{\lambda} \cdot \left(\frac{D_{k,\mu_j}^{\mathbf{M}}}{\lambda}\right)^\beta \cdot e^{-(D_{k,\mu_j}^{\mathbf{M}}/\lambda)^\beta} \\ \ln\left(\frac{D_{k,\mu_j}^{\mathbf{M}}}{\lambda}\right) \cdot \left(\frac{D_{k,\mu_j}^{\mathbf{M}}}{\lambda}\right)^\beta \cdot e^{-(D_{k,\mu_j}^{\mathbf{M}}/\lambda)^\beta} \end{bmatrix} \quad (9)$$

2.3. Open-Set Re-ID Framework

The framework for achieving re-ID is described in figure 1. Our framework has two major components, (a) pedestrian detection using state-of-the-art Mask-RCNN [19], and (b) Open to close set (Probe/Gallery) conversion via Weibull distribution before similarity ranking.

Pedestrian Detection using Mask-RCNN. Various pedestrian detectors [19, 20] have evolved featuring “proposal + CNN” approach. Among them Mask-RCNN [19] yield most accurate localization compared to other detectors. The backbone of Mask-RCNN is CNN + Feature Pyramid Network (FPN) [21] that pools pedestrian ROIs combining feature maps of various CNN layers. The pooled regions are further utilized to generate bounding boxes and feature vectors for Open-Set metric learning.

Open-Set Re-ID via Weibull rejection. A Weibull distribution (parameterized by $\mathbf{w}$) is learned along with metric $\mathbf{M}$ (see section 2.2) over generated feature. The learned distribution assigns a low probability value to dissimilar pairs in new metric space. This fact is utilized to reject some irrelevant samples from probe set before re-ID ranking over the dynamically generated gallery list. If a probe ID is rejected by all the gallery samples of a gallery list in a frame, then the probe is inferred to be absent (in that frame). Similarity rankings are then performed only with the remaining probe samples.

3. EXPERIMENTS

In this section, we discuss about the dataset, evaluation measures and Implementation details. We then present a detailed comparison with different feature extractors and metric learning methods.

3.1. Dataset Description

We have adopted PRW [6] dataset for evaluation. The dataset consists of 11,816 frames taken from 6 different views with 5 views having a resolution of 1080×1920 and the remaining view having a resolution of 576×720 . Pedestrian annotations at every 25th frame are available. Out of total 43,100 annotated pedestrians, 34,304 are assigned ID ranging from

1 to 932 and rest ambiguous ones are assigned -2. In most re-ID datasets only cropped ROIs of pedestrians are available which restricts evaluation of our model (detection + re-identification) on them. So, we have to limit our experiment to the PRW dataset only where full frames are available. Please note that since re-identification is generally performed over non-overlapping views (for probe and gallery sets) or overlapping views with sufficient temporal discontinuity the temporal relationship between two consecutive frames can be ignored.

3.2. Evaluation Measures

We have modified two measures [5], namely, Detection and Identification Rate (DIR) and False Accept Rate (FAR) for evaluating our OS-LMNN model. Usually, ROC is obtained by varying threshold τ over distance based similarity. We instead vary τ over Weibull distribution (ρ) as:

$$\begin{aligned} DIR(\tau, k) &= \frac{|\{p : p \in P_G, \text{rank}(p) \leq k, \rho(D_{pg}^{\mathbf{M}}) \geq \tau\}|}{|P_G|} \\ FAR(\tau) &= \frac{|\{p : p \in P_N, \text{and } \rho(D_{pg}^{\mathbf{M}}) \geq \tau\}|}{|P_N|} \end{aligned} \quad (10)$$

where, P_G, P_N are the two probe sets and G is the gallery with $g \in G$. Here, gallery set G consists of person common to set P_G but not in set P_N . Cumulative Matching Characteristics (CMC) curves are then constructed from rank-k recognition rate. Please note that DIR vs. FAR becomes CMC, when $FAR = 100\%$. For having fair comparisons with other metric learning methods, we have normalized distance between any two samples to $[0, 1]$ based on the maximum pairwise distance before using equation (10).

3.3. Implementation Details

All the implementations are done in python v2.7 using open-source libraries scipy, sklearn. Mask-RCNN [19] (available at <https://github.com/facebookresearch/Detectron>) is pre-trained in keras API using tensorflow in background over PRW dataset as pedestrian detector. Our framework has two hyper-parameters, namely, (a) margin α (equation 5) and (b) regularization constant γ (equation 6) which are experimentally set to 25 and 0.5 respectively.

3.4. Performance Comparison

We compare our model performance with four feature descriptors and four metric learning methods for each of the two state-of-the-art pedestrian detectors, namely, DPM [8] and Mask-RCNN [19]. The feature descriptors includes Bag-of-Words (BoW) vector [7], HistLBP [22], LOMO [14] and gBiCov [23]. The metric learning methods used for comparison are LMNN [12], KISSME [13], DNS [15] and XQDA [14]. Please note that currently no deep learning methods in literature have addressed open-set person re-ID problem. Moreover, the performance evaluation criteria used in open-set re-ID problems (see equation 10) are different from that of the close-set re-ID task. Therefore, the results of close-set re-ID methods are not directly comparable with our

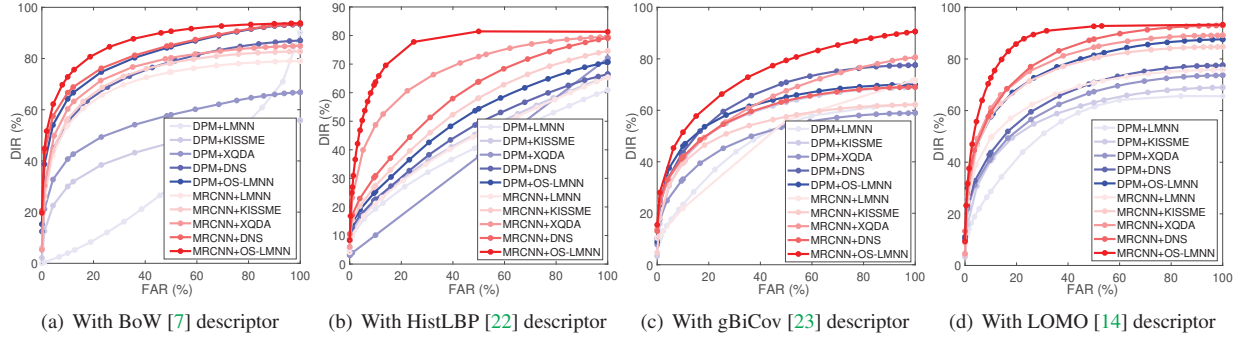


Fig. 2. ROC Curve with DIR vs FAR comparison at rank-1 recognition rate for different feature descriptors

proposed solution.

We first perform an exhaustive analysis with DPM as the pedestrian detector and the results are shown in table 1. Detected bounding boxes of DPM detectors have many false positives as compared to the Mask-RCNN detector. Yet the proposed metric learning model has performed well with respect to other metric learning methods. As can be seen from this table, OS-LMNN has outperformed various combinations of different feature descriptors and different metric learning methods by attaining a maximum DIR value of 38.69% at 1% FAR. The best competitor among other methods is BoW [7] + DNS [15] with a DIR of 30.03% at 1% FAR. This combination is also slightly lagging behind our 2nd best combination LOMO [14] + OS-LMNN (DIR = 30.04%). Also at other levels of FAR, specifically, at 10%, 50% and 100% we have secured highest performance with DIR of 64.23%, 84.62% and 93.34% respectively. Some metric learning methods drastically under-performed with DPM detectors as the influence of other false detection have overwhelmed over genuine pairs.

Mask-RCNN based detections are more accurate. This further increases the overall efficiency of our OS-LMNN metric learning. Please see table 2 in this connection. We have achieved a 6% performance boost with DIR of 44.82% for our best combination Mask-RCNN + BoW at 1% FAR. Among other methods DNS + BoW have performed best but with a lower DIR of 43.51% at 1% FAR. The improved performance of DNS is due to incorporation of non-linearity achieved through kernels. Still, for several combinations, the proposed OS-LMNN, in-spite of its linear nature, has clearly outperformed DNS.

The impact of joint learning of the Mahalanobis metric using LMNN with the Weibull distribution is evident from the ROC plots of figure 2. For all the cases, the LMNN method has performed miserably low as compared to our solution. For the DPM detector, the best descriptor combination with LMNN is LOMO with a DIR of 12.64% which is pretty low as compared to ours (30.04%) at 1% FAR. Though LMNN has improved for Mask-RCNN detector (DIR = 24.81%), it still considerably lags with respect to our OS-LMNN. This clearly shows the impact of Weibull based metric learning

especially when open-set rejection is concerned.

Table 1. DIR vs. varying FAR for Rank-1 scores with the DPM detector. Best values are shown in **bold**.

Detector	Feature	Recognizer	FAR(%)				AUC (%)
			1	10	50	100	
DPM [8]	HistLBP [22]	LMNN [12]	9.89	19.34	41.21	60.92	39.64
		KISSME [13]	11.17	21.83	46.01	65.45	43.88
		DNS [15]	12.79	22.70	49.34	66.50	45.98
		XQDA [14]	3.92	10.12	37.65	72.06	37.65
		OS-LMNN (Ours)	14.31	25.40	54.37	70.71	50.26
		OS-LMNN (Ours)	14.31	25.40	54.37	70.71	50.26
	LOMO [14]	LMNN [12]	12.64	28.69	61.36	65.58	53.89
		KISSME [13]	15.95	40.60	63.11	68.98	58.34
		DNS [15]	23.12	43.72	70.87	77.58	65.22
		XQDA [14]	21.97	41.54	67.33	73.70	61.96
		OS-LMNN (Ours)	30.04	57.63	81.42	87.61	76.30
		OS-LMNN (Ours)	30.04	57.63	81.42	87.61	76.30
	BOW [7]	LMNN [12]	0.43	3.96	29.84	90.11	32.85
		KISSME [13]	12.56	30.04	47.07	55.82	44.35
		DNS [15]	30.03	55.30	78.71	87.03	74.19
		XQDA [14]	21.17	40.72	58.02	66.82	55.16
		OS-LMNN (Ours)	38.69	64.23	84.62	93.34	80.83
		OS-LMNN (Ours)	38.69	64.23	84.62	93.34	80.83
	gBiCov [23]	LMNN [12]	10.12	23.41	54.70	62.31	48.24
		KISSME [13]	15.95	40.60	63.11	68.98	58.34
		DNS [15]	23.12	43.72	70.87	77.58	65.22
		XQDA [14]	17.57	33.22	53.86	58.96	49.57
		OS-LMNN (Ours)	24.03	46.10	65.14	70.09	61.40
		OS-LMNN (Ours)	24.03	46.10	65.14	70.09	61.40

Table 2. DIR vs. varying FAR for Rank-1 scores with Mask-RCNN detector. Best values are shown in **bold**.

Detector	Feature	Recognizer	FAR(%)				AUC (%)
			1	10	50	100	
Mask-RCNN [19]	HistLBP [22]	LMNN [12]	10.08	21.23	45.54	65.36	43.23
		KISSME [13]	15.21	27.42	58.14	74.59	53.68
		DNS [15]	17.19	30.92	63.87	79.13	58.54
		XQDA [14]	23.23	48.72	72.64	79.54	67.61
		OS-LMNN (Ours)	26.92	63.71	81.45	81.51	76.11
		OS-LMNN (Ours)	26.92	63.71	81.45	81.51	76.11
	LOMO [14]	LMNN [12]	24.81	49.45	70.79	75.9	65.99
		KISSME [13]	26.63	57.95	80.3	84.73	74.86
		DNS [15]	31.96	59.41	87.68	93.04	81.29
		XQDA [14]	28.03	61	84.53	89.19	78.80
		OS-LMNN (Ours)	31.65	72.7	92.65	93.31	87.15
		OS-LMNN (Ours)	31.65	72.7	92.65	93.31	87.15
	BOW [7]	LMNN [12]	25.3	54	74.74	79.02	69.83
		KISSME [13]	26.25	56.41	78.27	82.78	73.10
		DNS [15]	43.51	66.68	85.25	93.48	81.85
		XQDA [14]	30.41	60.29	80.28	84.95	75.49
		OS-LMNN (Ours)	44.82	72.82	90.66	93.8	86.23
		OS-LMNN (Ours)	44.82	72.82	90.66	93.8	86.23
	gBiCov [23]	LMNN [12]	11.62	21.93	56.01	71.54	49.17
		KISSME [13]	18.05	39.39	57.48	62.18	53.58
		DNS [15]	24.88	41.9	63.69	69.06	59.13
		XQDA [14]	24.06	41.72	67.57	80.6	63.50
		OS-LMNN (Ours)	28.04	51.49	79.4	90.61	74.24
		OS-LMNN (Ours)	28.04	51.49	79.4	90.61	74.24

4. CONCLUSION

In this paper, we have proposed a new metric learning method especially for performing open-set re-ID in wild. We introduced the concept of Weibull distribution for improved metric learning that better enforces imposter separation while training. In future, we plan to extend the proposed open set metric learning framework to deep learning based approaches for re-ID problems.

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